

Weekly Review and Practice Practical

Task 1 :- On Cisco Packet Tracer, create a network with three switches and assign at least two different VLANs (e.g., VLAN 10 for 'Sales', VLAN 20 for 'Support'). Connect two PCs to each switch and assign each PC to the correct VLAN based on department.

ANS :-

1. Create VLANs on All 3 Switches (Switch1, Switch2, Switch3) :

```
enable
configure terminal
vlan 10
name Sales
vlan 20
name Support
```

2. Connect PCs & Assign Ports on Each Switch :

- **PC1 (Sales): Connect to Fa0/1**

```
interface fa0/1
switchport mode access
switchport access vlan 10
```

PC2 (Support) : Connect to Fa0/2

```
interface fa0/2
switchport mode access
switchport access vlan 20
```

3.Enable Trunking Between Switches :

- On inter-switch link ports (e.g., Fa0/23, Fa0/24) :

```
interface range fa0/23 - 24
switchport mode trunk
```

Task 2 :- Configure trunk ports between the switches to allow VLAN traffic to pass through. Verify trunking by using the show interfaces trunk command and submit a screenshot of the output.

ANS :-

1. **Configure Trunking on Inter-Switch Ports :** On each switch CLI, enter configuration mode for the connected links (e.g., ports Fa0/23 and Fa0/24) :

```
enable
configure terminal
interface range fa0/23 - 24
switchport mode trunk
end
```

2. **Verify Trunking** : Run this command in Privileged EXEC mode (# prompt) :

```
show interfaces trunk
```

3. **Output Screenshot** :



```
Switch#show interfaces GigabitEthernet 0/1 switchport
Name: Gig0/1
Switchport: Enabled
Administrative Mode: trunk
Operational Mode: down
Administrative Trunking Encapsulation: dot1q
Operational Trunking Encapsulation: dot1q
Negotiation of Trunking: On
Access Mode VLAN: 1 (default)
Trunking Native Mode VLAN: 1 (default)
Voice VLAN: none
Administrative private-vlan host-association: none
Administrative private-vlan mapping: none
Administrative private-vlan trunk native VLAN: none
Administrative private-vlan trunk encapsulation: dot1q
Administrative private-vlan trunk normal VLANs: none
Administrative private-vlan trunk private VLANs: none
Operational private-vlan: none
Trunking VLANs Enabled: ALL
Pruning VLANs Enabled: 2-1001
Capture Mode Disabled
Capture Mode Allowed: ALL
Protected: false
--More--
```

Task 3 :- Set up a Layer 3 device (Router-on-a-Stick using a router or a multilayer switch) to enable inter-VLAN routing between VLAN 10 and VLAN 20. Assign appropriate sub-interfaces and IP addresses for each VLAN.

ANS :-

1. Connect & Trunk Link : Connect switch port Fa0/24 to router port Gi0/0/0.

• **On Switch CLI :**

```
enable
configure terminal
interface fa0/24
switchport mode trunk
```

2. Configure Sub-interfaces on Router CLI :

```
enable
configure terminal
interface gi0/0/0
no shutdown

interface gi0/0/0.10
encapsulation dot1Q 10
ip address 192.168.10.1 255.255.255.0

interface gi0/0/0.20
encapsulation dot1Q 20
ip address 192.168.20.1 255.255.255.0
```

3. Set PC Gateways :

- VLAN 10 PCs (Sales): Gateway = 192.168.10.1
- VLAN 20 PCs (Support): Gateway = 192.168.20.1

4. **Verify** : Ping from a VLAN 10 PC to a VLAN 20 PC → Success!

Task 4 :- Test connectivity by pinging between a PC in VLAN 10 and a PC in VLAN 20. If pings fail, troubleshoot and fix the configuration so that inter-VLAN communication works.

ANS :-

1. **Test Ping** : Open Command Prompt on Sales PC (VLAN 10) and run :

```
ping 192.168.20.2
```

2. **If Ping Fails** — Check & Fix These 3 Common Mistakes :

- PC IP & Gateway : Check PC network settings.
 - VLAN 10 PC Gateway must be 192.168.10.1.
 - VLAN 20 PC Gateway must be 192.168.20.1.

- Switch Trunk Port : Ensure switch port connected to router (Fa0/24) is set to trunk :

```
interface fa0/24
switchport mode trunk
```

Router Interface: Ensure main router port is enabled :

```
interface gi0/0/0
no shutdown
```

3. **Verify Fix:** Re-run ping 192.168.20.2 → Success!